

SOUTH JERSEY, NJ, USA

WMO: 724074

Lat: 39.941N

Lon: 74.841W

Elev: 15

StdP: 101.15

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 93780

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.4	(c) -8.9	(d) -20.9	(e) 0.6	(f) -9.9	(g) -18.0	(h) 0.8	(i) -7.1	(j) 9.8	(k) 2.4	(l) 8.9	(m) 0.2	(n) 3.1	(o) 290	(p) 0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	33.6	23.6	32.3	23.1	31.0	22.6	25.4	30.6	24.7	29.4	24.0	28.3	3.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	18.9	27.2	23.2	18.0	26.5	22.7	17.4	26.1	77.5	30.4	74.6	29.3	72.0	28.4	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.3	7.3	6.3	-14.4	35.7	2.6	1.7	-16.3	36.9	-17.8	38.0	-19.3	38.9	-21.2	40.2
			-15.0	26.9	2.6	0.7	-16.8	27.4	-18.4	27.8	-19.8	28.1	-21.7	28.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.7	0.2	1.5	5.9	11.8	17.2	21.9	24.6	23.6	19.9	13.6	7.9	3.5	(6)	
(7)		DBStd	9.51	5.73	5.10	5.01	4.72	4.28	3.56	2.81	2.85	3.60	4.49	4.56	5.07	(7)	
(8)		HDD10.0	1048	307	242	146	31	1	0	0	0	0	16	93	213	(8)	
(9)		HDD18.3	2662	562	471	385	204	75	9	0	1	24	158	312	461	(9)	
(10)		CDD10.0	2036	4	4	19	86	223	358	452	422	298	127	32	10	(10)	
(11)		CDD18.3	607	0	0	0	8	39	118	193	165	72	12	0	0	(11)	
(12)		CDH23.3	5501	0	0	5	105	393	1089	1853	1422	562	71	1	0	(12)	
(13)		CDH26.7	1991	0	0	0	28	121	393	765	515	158	11	0	0	(13)	
(14)	Wind	WSAvg	2.4	3.0	3.0	3.3	3.0	2.2	2.0	1.8	1.5	1.7	2.2	2.5	2.7	(14)	
(15)	Precipitation	PrecAvg	1153	93	64	107	93	92	97	114	121	93	93	81	105	(15)	
(16)		PrecMax	1454	164	128	196	227	172	224	266	330	191	265	165	218	(16)	
(17)		PrecMin	869	37	13	26	26	31	36	43	31	10	14	24	28	(17)	
(18)		PrecStd	148	32	29	42	43	32	51	49	64	52	53	36	44	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.3	19.7	23.7	30.2	32.3	34.6	36.2	35.1	32.9	28.0	22.4	20.1	(19)	
(20)			MCWB	15.1	13.1	15.6	18.6	21.8	23.7	24.2	24.4	23.3	20.7	17.3	16.9	(20)	
(21)		2%	DB	15.3	16.0	20.1	26.5	30.1	32.5	34.0	32.8	30.5	25.6	19.9	17.0	(21)	
(22)			MCWB	12.8	11.5	13.5	16.8	21.0	23.1	23.8	23.8	22.2	19.8	15.3	14.6	(22)	
(23)		5%	DB	12.0	12.7	17.2	23.3	27.6	30.9	32.5	31.1	28.3	23.2	17.7	14.4	(23)	
(24)			MCWB	9.0	8.8	11.6	15.5	19.8	22.5	23.3	22.9	21.3	18.5	14.2	12.0	(24)	
(25)		10%	DB	8.8	9.4	14.2	20.7	25.4	28.9	31.0	29.5	26.7	21.3	15.9	11.5	(25)	
(26)			MCWB	6.0	5.8	9.8	14.3	18.6	21.2	22.9	22.1	20.6	17.3	12.6	9.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.9	16.0	16.9	20.4	23.3	25.2	26.5	26.4	25.1	23.0	18.5	17.5	(27)	
(28)			MCDWB	17.5	18.6	21.7	26.5	29.4	31.3	33.0	32.3	29.5	25.3	20.9	18.7	(28)	
(29)		2%	WB	13.0	12.7	14.6	18.5	22.2	24.4	25.4	25.4	24.1	21.3	16.9	15.1	(29)	
(30)			MCDWB	15.0	14.9	18.6	24.1	28.0	29.8	31.0	30.3	27.5	24.0	18.9	16.6	(30)	
(31)		5%	WB	9.4	8.9	12.6	16.9	21.1	23.6	24.7	24.6	23.1	19.8	15.1	12.6	(31)	
(32)			MCDWB	11.3	12.2	15.9	21.8	26.2	28.6	29.7	28.7	25.9	22.2	17.2	14.2	(32)	
(33)		10%	WB	6.5	6.2	10.1	15.2	19.8	22.7	24.0	23.8	22.2	17.9	13.2	9.5	(33)	
(34)			MCDWB	8.4	9.1	14.1	20.0	24.1	27.2	28.6	27.4	25.2	20.5	15.2	11.1	(34)	
(35)	Mean Daily Temperature Range			MDBR	8.6	9.3	10.2	11.6	11.3	10.8	10.7	10.5	11.1	10.5	10.3	8.6	(35)
(36)		5% DB	MCDBR	13.1	13.9	14.8	15.8	14.2	12.7	12.3	11.9	12.9	12.6	12.8	11.0	(36)	
(37)			MCWBR	10.8	10.5	9.6	8.0	6.7	5.2	4.3	4.5	5.9	7.2	8.9	10.0	(37)	
(38)		5% WB	MCDBR	12.1	13.0	13.5	13.4	12.4	11.1	10.4	9.7	10.0	10.0	10.5	10.7	(38)	
(39)			MCWBR	11.0	11.0	10.0	8.2	6.5	5.3	4.3	4.4	5.3	6.9	8.7	10.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.313	0.321	0.360	0.391	0.438	0.502	0.503	0.495	0.409	0.361	0.329	0.310	(40)	
(41)		taud		2.465	2.437	2.385	2.322	2.222	2.085	2.097	2.105	2.318	2.461	2.509	2.522	(41)	
(42)		Ebn at Noon		853	898	894	883	845	789	784	778	830	840	828	831	(42)	
(43)		Edh at Noon		79	93	109	124	140	161	158	153	116	90	75	70	(43)	
(44)	All-Sky Solar Radiation	RadAvg		1.94	2.85	3.77	4.87	5.41	5.93	6.05	5.25	4.35	3.10	2.21	1.69	(44)	
(45)		RadStd		0.17	0.31	0.31	0.42	0.63	0.49	0.29	0.34	0.45	0.32	0.20	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (71 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+122	+70		

Nomenclature: See separate page